

New curvilinear scheme for elastic wave propagation in presence of curved topography

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ABSTRACT

We study a new curvilinear scheme for wave propagation modelling in presence of topography. The discrete scheme takes advantage of recent developments in aeroacoustics. Our new scheme relies on the conventional grid coupled with optimized filters to remove numerical noise in case of strong material heterogeneity. We used non-centred stencils for free surface implementation and optimized the explicit Runge-Kutta scheme for the time differencing. We performed a complete theoretical stability and dispersion analysis of the discrete scheme. Finally, we illustrate the numerical accuracy of the new scheme by intensive experiments.

Key words: Computing aspects, Modelling, Topography.

INTRODUCTION

With the increasing scarcity of oil, the oil companies explore ever more challenging geological features. The complexity can arise because of intricate geological structures (e.g., salt ridges, weathered zones, foothills). In the particular context of foothills, specific elastic wave propagation phenomena appear from the presence of topography: scattering of waves, site effects caused by interferences that cause amplification of the wave amplitude and complex propagation of ground-roll, including mode conversions between surface and body waves.

An accurate modelling in presence of topography requires a correct description of the free surface. Generally, finite element and finite volume methods are the most suitable approaches to handle accurately the free surface. For instance, Komatitsch and Tromp (1999) introduced a spectral element scheme as a computationally efficient way of tackling this problem. The method is based on high-order variational methods that use Gauss-Lobatto-Legendre nodes. The quadrature rules for the approximation of mass and stiffness matrices lead to a diagonal matrix. Thus, the matrix inversion step is avoided and performance is preserved. Another approach

is based on the discontinuous Galerkin method (Falk and Richter 1999; Monk and Richter 2005). The main idea of the method consists in using discontinuous functions and enforces the continuity weakly across interfaces by adding extra terms. The resulting mass matrix is block diagonal and its inversion is straightforward. Recently, new interest has been raised on the finite volume method (LeVeque 2002; Käser and Iske 2005). Brossier, Virieux and Operto (2008) applied the so-called *P0* finite volume scheme (the unknown functions are supposed to be constant on the triangular mesh nodes) in the frequency domain, they pointed out the good potential of the method particularly for the inversion purpose.

All these methods are well adapted to handle efficiently a curved free surface. However, operational projects, such as seismic feasibility studies, require propagating elastic wavefields accurately over extensive and finely sampled models and imply huge and often prohibitive computing power. In this context, the finite difference method still remains the most appropriate technique to use. This technique offers an attractive trade-off between accuracy and computational efficiency. However, in presence of topography, staircase approximation of the free surface may hinder the accuracy of the Cartesian scheme. Lombard *et al.* (2008) developed an efficient method to avoid spurious diffractions induced by the staircase

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description of boundaries. Indeed, the contribution of the topography is computed and then injected into the numerical scheme near the boundaries. The irregularity of the mesh resulting from this approach may cause implementation issues. In particular, for large parallel computing platforms this leads to complex load balance problems for large 3D simulations. Another approach to overcome the inaccuracy of the staircase approximation within Cartesian finite differences is curvilinear coordinates (Fornberg 1988b). The main idea is to transform the initial Cartesian model into a new computing model having a flat topography. The grid regularity (i.e., rectangular model) preserves then the flexibility and the parallel performance. Hestholm and Ruud (1994) adapted the curvilinear coordinates to finite differences. Hence, they expressed the new free surface condition since the earlier work was not directly transposed to finite differences.

In this work we study a new discrete scheme based on curvilinear finite differences. The scheme relies on the use of a conventional grid coupled with optimized filters as usually considered in aeroacoustics. In the first section we review the main equations and we establish the theoretical framework. We then introduce the discrete scheme for both internal and boundary points and we analyse its stability, thanks to the new curvilinear stability condition. We end up with three meaningful tests that emphasize the good accuracy of this approach.

THEORETICAL BACKGROUND

In the foothills context, only the top edge of the geological model is curved. Assuming that $(z_0(x), x \in [0, x_{\max}])$ is the equation of the topography, so that $|z_0(x)|$ remains smaller than the maximum depth H of the geological model. Follow-

ing Fornberg (1988b), we introduce the coordinate transformation that we use in this work

$$\begin{cases} x = \xi, \\ z = z_0 + \frac{H - z_0}{H} \eta, \end{cases} \implies \begin{cases} \xi = x, \\ \eta = \frac{H(z - z_0)}{H - z_0}. \end{cases} \quad (1)$$

Substituting the new spatial derivatives in the 2D system of elastodynamics with respect to the Jacobian matrix and assuming the differentiability of the topography curve z_0 (Tessmer, Kosloff and Behle 1992), we get

$$\begin{cases} \rho \frac{\partial v_x}{\partial t} = \frac{\partial \sigma_{xx}}{\partial \xi} + a \frac{\partial \sigma_{xx}}{\partial \eta} + b \frac{\partial \sigma_{xz}}{\partial \eta}, \\ \rho \frac{\partial v_z}{\partial t} = \frac{\partial \sigma_{xz}}{\partial \xi} + a \frac{\partial \sigma_{xz}}{\partial \eta} + b \frac{\partial \sigma_{zz}}{\partial \eta}, \\ \frac{\partial \sigma_{xx}}{\partial t} = (\lambda + 2\mu) \frac{\partial v_x}{\partial \xi} + (\lambda + 2\mu)a \frac{\partial v_x}{\partial \eta} + \lambda b \frac{\partial v_z}{\partial \eta}, \\ \frac{\partial \sigma_{zz}}{\partial t} = \lambda \frac{\partial v_x}{\partial \xi} + \lambda a \frac{\partial v_x}{\partial \eta} + (\lambda + 2\mu)b \frac{\partial v_z}{\partial \eta} \\ \frac{\partial \sigma_{xz}}{\partial t} = \mu \left(\frac{\partial v_z}{\partial \eta} + a \frac{\partial v_z}{\partial \eta} + b \frac{\partial v_x}{\partial \eta} \right). \end{cases} \quad (2)$$

For the sake of simplicity of exposure the source function has been omitted. The velocity components are (v_x, v_z) , $(\sigma_{xx}, \sigma_{xz}, \sigma_{zz})$ the stress tensor components, (λ, μ) the Lamé's parameters and (a, b) the metrics of the transformation, which are defined by

$$\begin{cases} b(\xi) = \frac{H}{H - z_0(\xi)}, \\ a(\xi, \eta) = -\frac{H z'_0(\xi)}{H - z_0(\xi)} \left(1 - \frac{\eta}{H} \right). \end{cases} \quad (3)$$

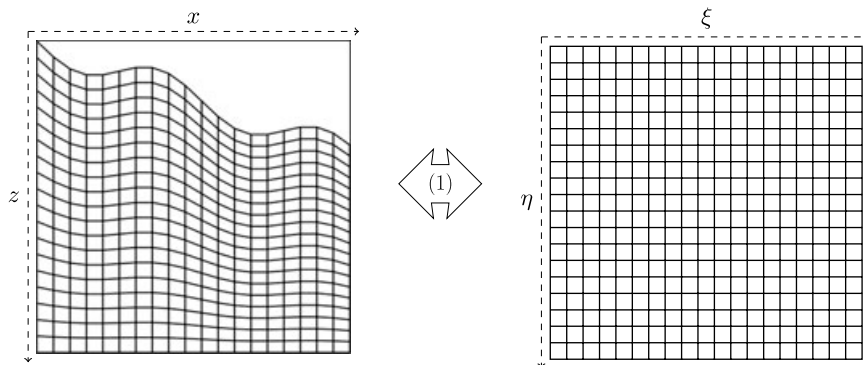


Figure 1 The initial Cartesian model (left) and the computation model (right) result in an application of the curvilinear coordinate transformation defined by equation (1).

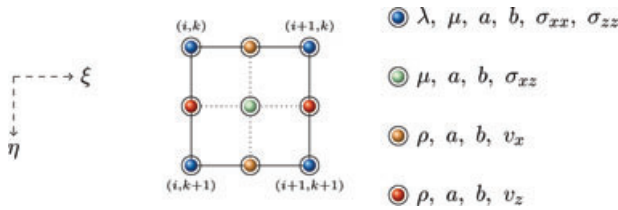


Figure 2 Staggered grid mesh for the 2D velocity-stress system in curvilinear coordinates. The metrics (a,b) are required at each grid point.

DISCRETE FINITE DIFFERENCE STENCIL

Finite difference grid

We developed a new curvilinear finite difference scheme based on a conventional grid. While the staggered and the rotated staggered grids are mostly used for the Cartesian case, their application is no longer suited to our problem. The main reasons are the following:

- Hestholm and Ruud (1994) extended the Cartesian staggered grid formulation (Virieux 1986) to curvilinear coordinates (see Fig. 2). Its application to curvilinear coordinates requires an extensive mean of computations. This prohibitive cost is mainly due to the computation of the spatial vertical derivatives and the interpolation operator needed to make the calculation feasible at the desired points. Additionally, the use of an interpolation operator can decrease the overall accuracy of the scheme as quoted by Magnier, Mora and Tarantola (1994) and Komatitsch, Coutel and Mora (1996).

Hestholm and Ruud in their different papers (Hestholm and Ruud 1994, 1998; Hestholm 1999; Hestholm and Ruud 2000; Ruud and Hestholm 2001; Hestholm 2003) never mentioned what they effectively used as an interpolation operator, which in general must implement a trade-off between the accuracy and the computational complexity of the scheme. The basic linear operator I_4 that has low accuracy in space (only 2nd order) is usually considered (see Fig. 3).

To summarize the computational complexity of the staggered curvilinear scheme: at each time step, updating one unknown function requires six computations of the spatial derivatives as well as the interpolation. Because of its prohibitive extra computational cost and diminishing accuracy, the staggered grid is not well suited to curvilinear coordinates, particularly, for large 3D models.

- Similar to the Cartesian coordinate framework, we can adapt the rotated staggered grid to curvilinear coordinates.

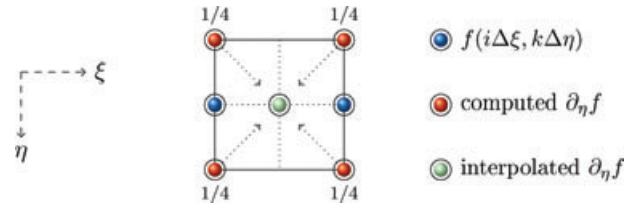


Figure 3 The computation of vertical derivatives requires interpolation at the desired point. Unfortunately, a basic interpolation degrades the accuracy of the numerical scheme.

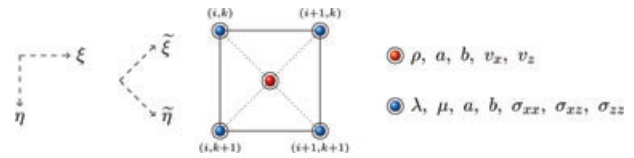


Figure 4 Rotated staggered grid mesh for the 2D velocity-stress system in curvilinear coordinates. The spatial derivatives are computed along the square axes ($\tilde{\xi}$, $\tilde{\eta}$).

In Fig. 4, we display a sketch of the considered grid. The rotated curvilinear scheme allows the computation of all the spatial derivatives without needing any interpolation operator. The spatial accuracy of the discrete scheme is then preserved and the computational complexity of the spatial derivatives remains similar to the Cartesian case. In this respect, the modified rotated staggered scheme can lead to a significant saving in computing time. However, according to Bohlen and Saenger (2006), the stress components on the free surface have to be set on the topography. This may lead in our case to an inconsistent scheme. Actually, our free surface condition describes the normal derivatives of the velocity in function of its horizontal ones. Admitting that the velocity components starts at $\Delta\eta/2$, the free surface condition cannot be satisfied directly. Furthermore, the source implementation within the rotated staggered grid requires a spatial smoothing (Hustedt, Operto and Virieux 2004), consequently it is not adapted for real amplitude analysis.

Our approach is based on the conventional grid formulation sketched in Fig. 5. The conventional grid was also used by Zhang and Chen (2006) and Appelö and Petersson (2009) for their problems in curvilinear coordinates. In particular, Zhang and Chen (2006) used a Mac-Cormack scheme for the discrete scheme implementation. Whereas, in our study, we used a high-order optimized scheme in both space and time for internal point treatment combined with new optimized non-centred stencils directly adapted from aeroacoustics.

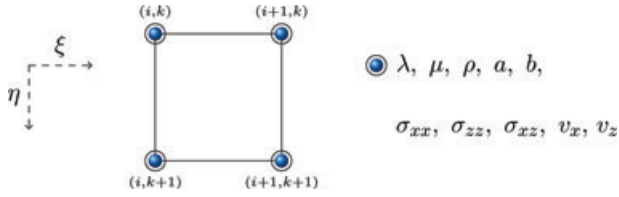


Figure 5 Conventional grid mesh for the 2D velocity-stress system in curvilinear coordinates. All the quantities are defined at the same grid position.

High-order spatial differencing

It is well-known that high-order discretization is required to avoid excessively fine meshes for many practical problems. In consequence, many high-order differencing methods have been developed for problems involving wave phenomena (Alford, Kelly and Boore 1974; Dablain 1986; Cohen and Joly 1990).

In the following we present different strategies for the design of a new high order based on a conventional grid. We first review and compare different stencils in term of their spectral accuracy, which lead us to define the stencil used in this work. First of all, we consider a function f and let Δx denote the spacing of a uniform grid. The discrete finite difference approximation $\frac{\partial f}{\partial x}$ of the spatial derivative $\frac{\partial f}{\partial x}$ at a grid point x_0 , consists of approximating $\frac{\partial f}{\partial x}$ by a weighted summation of the function values at some neighbouring grid points. For a centred $(2N+1)$ -point stencil with antisymmetric coefficients a_j , the approximation reads

$$\frac{\partial f}{\partial x}(x_0) = \frac{1}{\Delta x} \sum_{j=1}^N a_j (f(x_0 + j\Delta x) - f(x_0 - j\Delta x)), \quad (4)$$

with the criterion:

$$\left| \frac{\partial f}{\partial x}(x_0) - \frac{\partial f}{\partial x}(x_0) \right| \leq M(\Delta x)^N, \quad (5)$$

where M is a positive constant independent of Δx .

Here, the stencil coefficients a_j are usually chosen antisymmetric in order to ensure a minimal dissipation.

Fourier analysis is required to analyse the accuracy of the difference formula. The Fourier basis function e^{ikx} is an eigenfunction of both the differentiation operator and also of all possible difference formulas. The accuracy of a given scheme can therefore be assessed merely by comparing the eigenvalues (Boyd 2006). Assuming that $\hat{f}$ is the spatial Fourier transform of f , we have

$$ik\hat{f}(x_0) \approx i\tilde{k}\hat{f}(x_0) = \frac{2i}{\Delta x} \left(\sum_{j=1}^N a_j \sin(jk\Delta x) \right) \hat{f}(x_0), \quad (6)$$

k is the physical (or exact) wavenumber and $\tilde{k}$ is the effective wavenumber of the approximating operator. We obtain

$$k\Delta x \approx \tilde{k}\Delta x = 2 \left(\sum_{j=1}^N a_j \sin(jk\Delta x) \right). \quad (7)$$

We note that for a staggered grid mesh, the previous equation writes:

$$k\Delta x \approx \tilde{k}\Delta x = 2 \left(\sum_{j=1}^N a_j \sin((j - \frac{1}{2})k\Delta x) \right). \quad (8)$$

The classical approach to derive the stencil weights a_j relies on the Taylor series truncation method (Fornberg 1988a) whereas the Taylor criterion (equation (5)) defines the spatial order. The Taylor criterion can be seen as a low-frequency relation. In fact, we have no information on the discretization accuracy in a range of frequencies but only an approximation of a function when $(\Delta x \rightarrow 0)$ (Rodrigues 1993).

Another interesting approach suggests lowering the order in the stencil construction to produce a scheme that includes a dispersive perspective (Haras and Taasan 1994; Zingg, Lomax and Jurgens 1996; Hu, Hussaini and Mantney 1996). We refer to such schemes as optimized schemes. Vichnevetsky and Schutter (1975) first proposed this methodology and Holberg (1987) studied the idea and extended it to linear wave equations. His approach is based on the definition of a new accuracy criterion. The coefficients a_j are then chosen optimally to minimize an error (ε_b^i) related to the group velocity

$$\varepsilon_b^i = \frac{\partial k}{\partial \tilde{k}} - 1. \quad (9)$$

Kindelan, Kamel and Sguazzero (1990) and Kamel, Sguazzero and Kindelan (1995) followed Holberg's methodology. Accordingly, they simplified the computation and improved the accuracy of Holberg's differentiators on the staggered grid formulation. In geophysics, the conventional grid has been neglected to the detriment of the staggered grid. Thus, the development of optimized stencils on the conventional grid is practically nonexistent. In computational aeroacoustics where the conventional grid is still relevant, Tam and Webb (1993) followed the same theory for optimized stencils construction. Thus, they introduced a new integral error ε_t^i based on dispersion analysis and defined by

$$\varepsilon_t^i = \int_{(k\Delta x)_l}^{(k\Delta x)_h} |k\Delta x - \tilde{k}\Delta x|^2 d(k\Delta x). \quad (10)$$

The coefficients a_j are then optimized to preserve minimal dispersion in a wavenumber range $[(k\Delta x)_l, (k\Delta x)_h]$ and to ensure a 4th order accurate stencil. The result was a

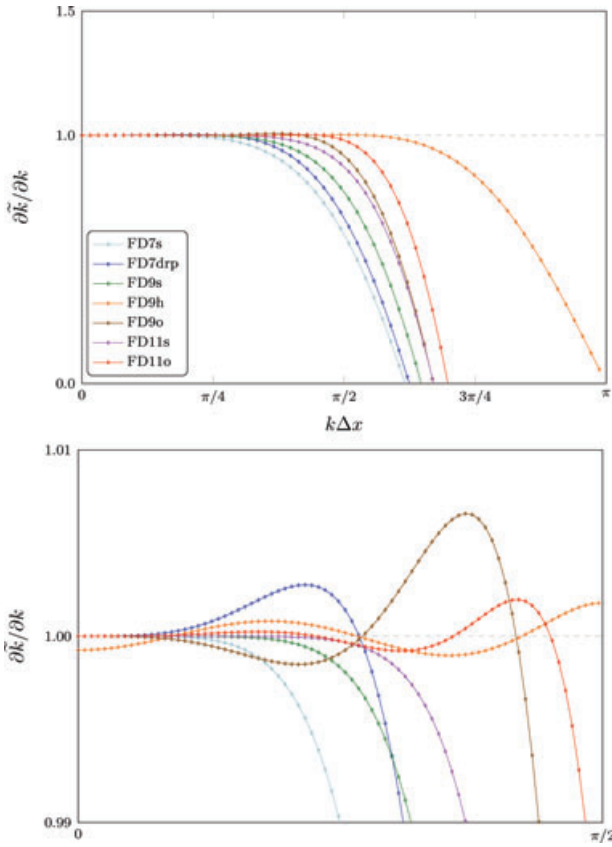


Figure 6 Dispersion-error characteristics of various centred stencils and a zoom on the area of interest. The optimized stencils DRP 7-point (FD7drp), Bogey 9-point (FD9o) and Bogey 11-point (FD11o) show better spectral accuracy than those based on the Taylor expansion. Although, the Taylor stencils are more accurate for low frequencies, the optimized stencils distribute the error over a large bandwidth. As expected, the Holberg 9-point stencil FD9h results in better resolution, since it is applied on a staggered grid.

7-point stencil, which is commonly called dispersion relation preserving.

Later on, Bogey and Bailly (2004) introduced a new relative error ε_b^i in order to minimize the relative difference, which is defined by

$$\varepsilon_b^i = \int_{\ln(k\Delta x)_l}^{\ln(k\Delta x)_h} |k\Delta x - \tilde{k}\Delta x| d(\ln(k\Delta x)). \quad (11)$$

Consequently, they developed several optimized stencils for various wavenumber ranges that are all 4th order accurate.

To summarize we plot in Fig. 6 the dispersion error characteristics of various centred stencils discussed previously. Particularly, we compare the standard stencils based on Taylor expansion, 7-point (FD7s) and 9-point (FD9s); DRP 7-point stencil (FD7drp) based on the work of Tam and Webb (1993) and the two Bogey stencils: 9-point (FD9o) and 11-

Table 1 Minimum number of points per wave length required to ensure a maximum error of E on the group velocity. As expected, the Holberg's stencil presents the best accuracy and it is applied on staggered grid

E	FD7s	FD7drp	FD9s	FD9h	FD9o	FD11s	FD11o
0.5%	9.02	6.80	6.92	3.47	5.93	5.84	4.24
1%	8.00	6.44	6.29	3.36	4.54	5.41	4.13

point (FD11o) (Bogey and Bailly 2004), to be complete we depict the Holberg 9-point stencil (FD9h) on a staggered grid.

Assuming a maximum error of $E = 5 \times 10^{-3}$ and $E = 10^{-2}$ on the group velocity, we report in Table 1 the required number of points per wavelength for each studied stencil.

According to these comparisons, the 11-point (FD11o) stencil presents the best spectral accuracy on the conventional grid. Therefore, we consider this stencil to update the internal points in space. As in aeroacoustics, we used optimized filters (Tam, Webb and Dong 1993) to remove spurious noise that arise for long time simulation or in presence of strong material heterogeneity. These high-order modes result in general on grid-to-grid oscillations that are not accurately calculated by the discretization algorithms (Vichnevetsky and Bowles 1982).

Free surface implementation

The free surface implementation is the key and the main difficulty of curvilinear finite differences. Consequently it has to be considered carefully. Zhang and Chen (2006) adapted the stress imaging technique to curvilinear coordinates. In this respect, they obtained the common form of the stress imaging in general coordinate systems. In our work, we used non-centred stencils initially introduced by Berland *et al.* (2007) for their boundary issues in aeroacoustics. This technique is very attractive as it is somehow equivalent to optimized adjusted finite differences approximations (Kristek, Moczo and Archuleta 2002) on a conventional grid and can be compared to Holberg improvements on Taylor stencils. Basically, the idea is to construct the coefficients a_j of non-centred stencils to minimize an integral error that includes the dissipation ($\text{Im}(\tilde{k}\Delta x)$) and the dispersion ($\text{Re}(\tilde{k}\Delta x)$) properties of the scheme. In the following we review the key ingredients of the stencil construction.

The non-centred derivative of a function f using M points on the left to x_0 and N points on the right writes

$$\frac{\tilde{\partial} f}{\partial x}(x_0) = \frac{1}{\Delta x} \sum_{j=-M}^N a_j f(x_0 + j\Delta x). \quad (12)$$

Applying a spatial Fourier transform to equation (12) reads

$$\tilde{k}\Delta x = -i \sum_{j=-M}^N a_j e^{ijk\Delta x}. \quad (13)$$

Berland *et al.* (2007) optimized the coefficients a_j in order to minimize a new dispersion/dissipation error ε_b^b defined by

$$\varepsilon_b^b = \int_{\ln(k\Delta x)_l}^{\ln(k\Delta x)_h} (1 - \alpha) |k\Delta x - \text{Re}(\tilde{k}\Delta x)| + \alpha |\text{Im}(\tilde{k}\Delta x)| d(\ln(k\Delta x)). \quad (14)$$

In the expression, α is a parameter chosen between 0–1 in order to control the balance of the two errors (dispersion and dissipation). For a wavenumber range between $\pi/16$ – $\pi/2$, they built a 4th order accurate 7-point and 11-point non-centred finite difference stencils. Figure 7 displays a sketch of the 11-point stencils in the vicinity of the boundary noted FDMNo for M points left and N points right, while FD11o denotes the centred stencil introduced in the previous section.

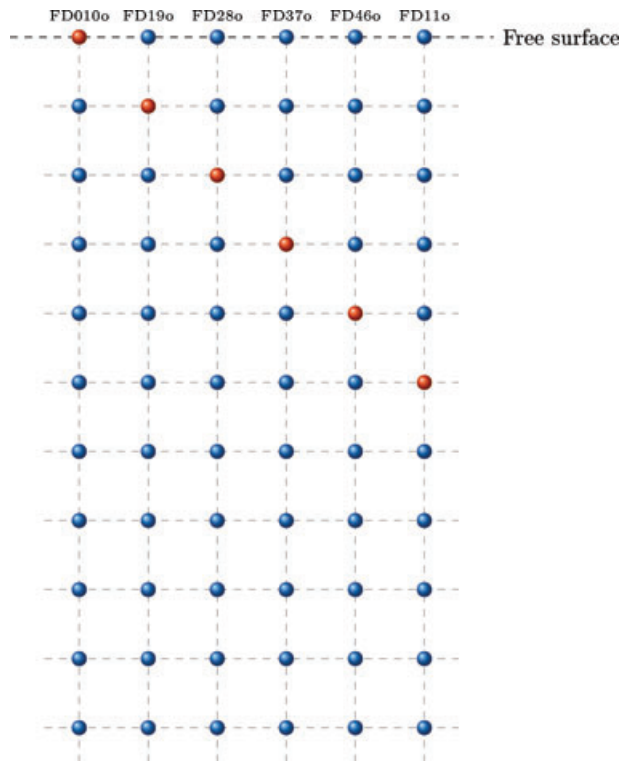


Figure 7 FDMNo, non-centred finite difference stencils for the free surface implementation. The computation of the normal derivatives at each grid position (red colour) is performed using the function values in the neighboring positions (blue colour). The centred stencil FD11o is used for internal points computation.

In Figs 8 and 9, we illustrate the dispersion and the dissipation of the 11-point non-centred stencils that we use in this work, while Table 2 reports the minimum number of points per wavelength required to ensure maximum errors of $E=5 \times 10^{-3}$ and $E=10^{-2}$ on both dispersion and dissipation of the discrete scheme.

Similar to the internal points treatment where we used a conventional grid, Berland *et al.* (2007) designed selective filters for the FDMNo stencils in order to avoid spurious modes. In this work, we integrate the 11-point non-centred stencils as well as their selective filters.

High-order time evolution

Among the different explicit schemes used for the elastic wave simulation, the most widely used one is probably the 2nd order Taylor stencil. However, for large-scale and long-time seismic modelling, the time discretization may produce severe dispersion and dissipation errors (Dablain 1986; Cohen and Joly 1990; Crase 1990; Chen 2006). For this purpose, various high-order schemes have been proposed to accurately approximate the time derivatives. For instance, Dablain (1986) proposed a scheme based on Taylor expansion for the second-order scalar wave equation, the idea is to replace the time derivatives by space derivatives using the Lax-Wendroff procedure. Following the same philosophy, Toro and Millington (2000) and Titarev and Toro (2005) extended the method based on Lax-Wendroff to 3D linear partial differential equations. They called it the ADER scheme for arbitrary high-order schemes using derivatives, since we can reach easily a high-order scheme in space and time. Alternately, Tal-Ezer (1986) developed a new scheme based on time extrapolation using Chebychev expansion. This method approximates the time derivative with a high accuracy, but the drawback lies on the extensive storage that it requires. Runge-Kutta algorithms have been also studied. Chen (2006) explored Nyström's method for the second-order scalar wave equation, the idea is to transform the wave equation to an equivalent first-order system on which he applied a Runge-Kutta scheme. In the velocity-stress formulation, the system is of first-order. Therefore, Runge-Kutta algorithms are well suited to design the time differencing.

Similar to the spatial differencing, most of Runge-Kutta algorithms use Taylor expansion to compute the coefficients of the temporal stencil. However, Zingg *et al.* (1996) and Wells and Renaut (1997) reported that high-order methods that combine optimized spatial and temporal stencils provide superior accuracy for long-range wave propagation.

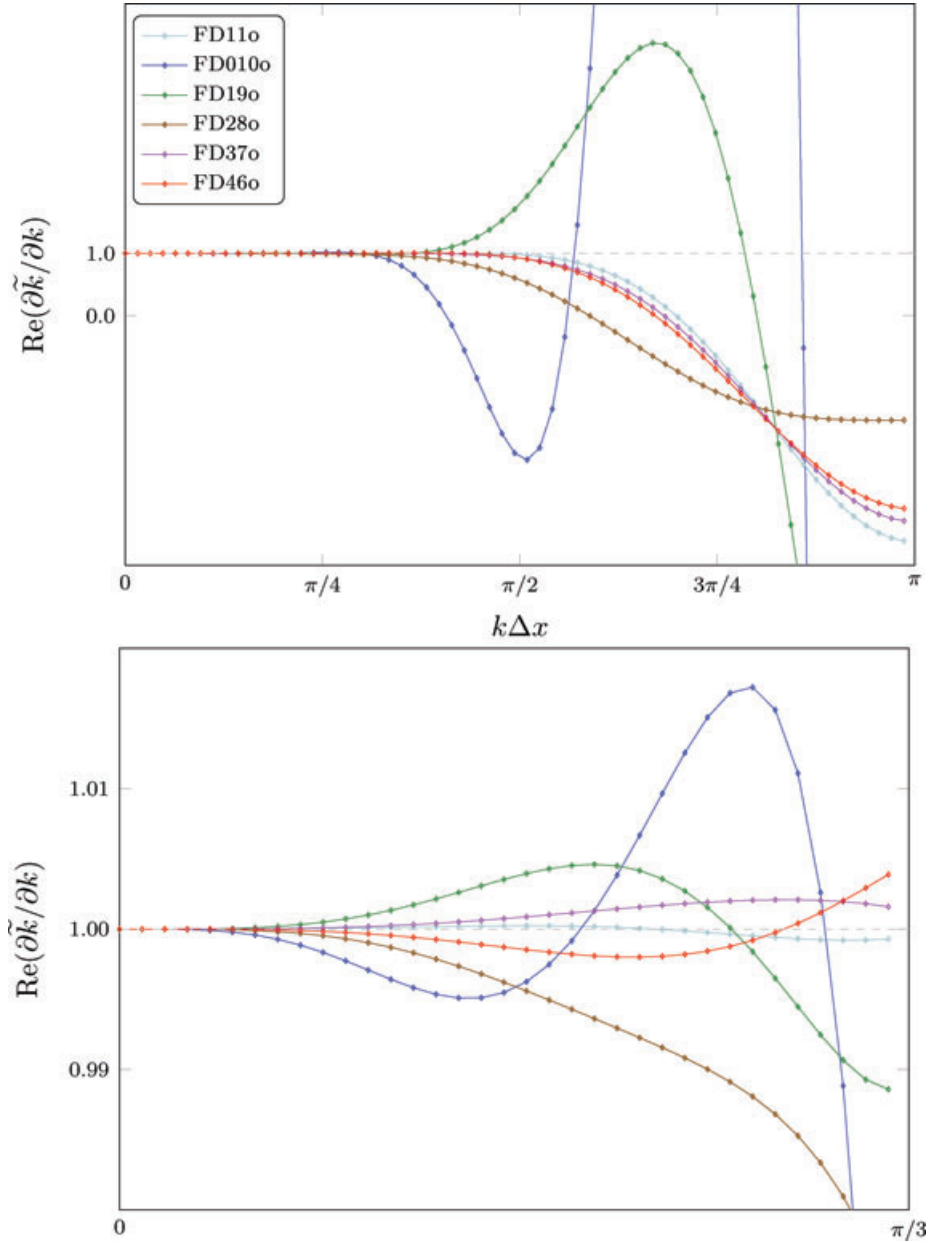


Figure 8 Spectral accuracy of the non-centred stencils that we use in work and a zoom in on the area of interest. Similar to the optimized centred stencils, the errors are distributed over a wider bandwidth.

We review hereunder the construction of an optimized Runge-Kutta scheme as usually used in computational aeroacoustics. Foremost, we consider the system of elastodynamics written in the conservation form,

$$\frac{\partial U}{\partial t} = S(U) = A_{\xi} \frac{\partial U}{\partial \xi} + A_{\eta} \frac{\partial U}{\partial \eta}, \quad (15)$$

For the sake of simplicity, we omit the source term. U is a vector containing the elastic wave components; stress and velocity. A_{ξ} and A_{η} are two matrices that depend on the petroelastic parameters and the metrics of the coordinate transformation.

We note $U^n = U(n\Delta t)$ and α_l the Runge-Kutta coefficients. The time evolution using an explicit p-stage Runge-Kutta

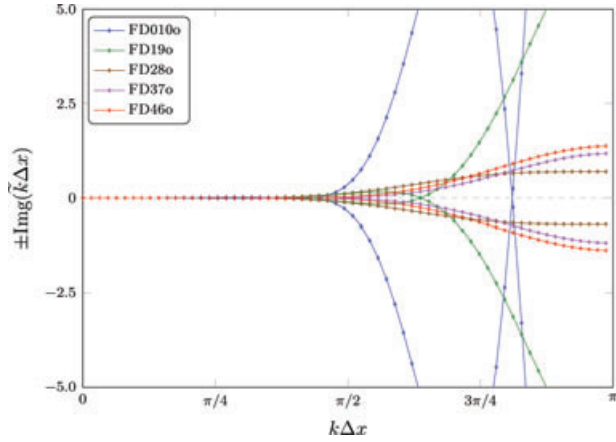


Figure 9 Dissipation-error characteristics the non-centered stencils considered in this work. The stencils are accurate over a large bandwidth of wavenumbers.

Table 2 Minimum number of points per wavelength allowed for a maximum error of E on the group velocity and on the scheme dissipation. For comparison, we report the values required for the internal stencil FD11o

E	FD010o	FD19o	FD28o	FD37o	FD46o	FD11o
0.5 %	10.13	7.03	11.06	5.04	5.95	4.23
1 %	8.67	6.45	8.04	4.80	4.48	4.13

algorithm advances the solution from $n\Delta t$ to $(n+1)\Delta t$ as:

$$\begin{cases} V^0 = U^n, \\ \vdots \\ V^\ell = U^n + \alpha_\ell \Delta t S(V^{\ell-1}) \quad \ell \in \{0, \dots, p\}, \\ \vdots \\ U^{n+1} = V^p. \end{cases} \quad (16)$$

In a compact form the algorithm reads

$$U^{n+1} = U^n + \sum_{j=1}^p \gamma_j (\Delta t)^j \frac{\partial^j U^n}{\partial t^j}, \quad \gamma_j = \prod_{\ell=p-j+1}^p \alpha_\ell. \quad (17)$$

Applying a temporal Fourier transform to equation (17), we derive the effective amplification factor G of the semi-discrete scheme

$$G(\omega \Delta t) = \frac{\hat{U}^{n+1}(\omega)}{\hat{U}^n(\omega)} = 1 + \sum_{j=1}^p \gamma_j (i\omega \Delta t)^j. \quad (18)$$

Here, $\hat{U}$ is the temporal Fourier transform of U . The effective angular frequency of scheme $\tilde{\omega}$ accounts for the phase of the amplification factor G , while the modulus of G represents the amplification rate. Thus, we can express both the disper-

sion of the scheme as $|\tilde{\omega} \Delta t - \omega \Delta t|$ and the dissipation as $1 - |G(\omega \Delta t)|$.

Hu *et al.* (1996) constructed the Runge-Kutta coefficients γ_j by minimizing an error ψ_b between the effective amplification factor G and the exact one $e^{i\omega \Delta t}$. The coefficients γ_j are then optimized in a frequency range $[(\omega \Delta t)_l, (\omega \Delta t)_h]$ to preserve a 2nd order accuracy

$$\psi_b = \int_{(\omega \Delta t)_l}^{(\omega \Delta t)_h} |G(\omega \Delta t) - e^{i\omega \Delta t}|^2 d(\omega \Delta t). \quad (19)$$

Bogey and Bailly (2004) followed the same philosophy, they constructed the coefficients γ_j in order to minimize the relative error ψ_b defined by

$$\psi_b = \int_{\ln(\omega \Delta t)_l}^{\ln(\omega \Delta t)_h} [(1 - G(\omega \Delta t)) + |\tilde{\omega} \Delta t - \omega \Delta t|/\pi] d(\ln(\omega \Delta t)) \quad (20)$$

In the next section we study the stability and the dispersion of the 6-stage Runge-Kutta algorithm developed by Bogey and Bailly (2004) and compare it to a classical 5-stage Runge-Kutta based on Taylor coefficients. In this study, our modelling scheme is based on the 6-stage optimized Runge-Kutta algorithm. The implementation of the discrete scheme is readily performed and its extension to curvilinear coordinates is straightforward.

Stability and dispersion analysis

We recall the effective amplification factor G of the semi-discrete scheme

$$G(\omega \Delta t) = \frac{\hat{U}^{n+1}(\omega)}{\hat{U}^n(\omega)} = 1 + \sum_{j=1}^p \gamma_j (i\omega \Delta t)^j. \quad (21)$$

A sufficient condition of stability is that $|G| < 1$ (Strikwerda 2004), this condition ensures that $U^n = G^n U^0$ remains bounded.

In Fig. 10 we depict the amplification factor of our integration algorithm (Bogey 6-stage). For comparison we also report the amplification factor of the 5-stage Taylor scheme.

Assuming an error of 2.5×10^{-3} (i.e., 0.25%) on the amplification factor, the values of $\omega \Delta t$ that satisfy the stability condition are reported in Table 3.

In the continuous equation (15), the angular frequency ($i\omega$) accounts for the eigenvalues of the Fourier transform $\hat{S}$ of the operator S . Therefore, to establish the stability condition we need to provide an upper-bound $|\Lambda_{max}|$ of the eigenvalues of $\hat{S}$. The Von Neumann method or plane wave analysis is often used to address the stability issues of partial differential equations. In the particular context of wave simulation, the idea is

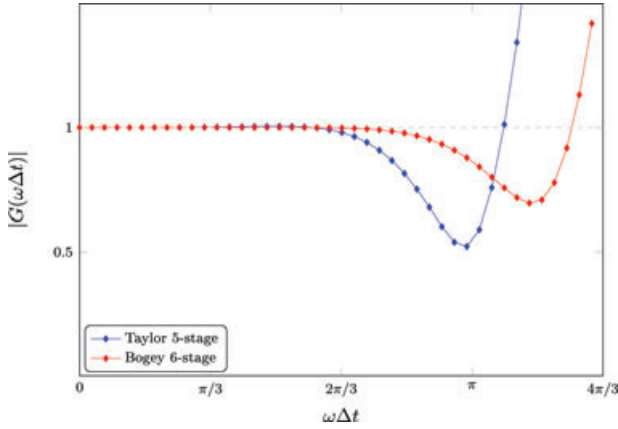


Figure 10 Comparative study of the amplification factor of different Runge-Kutta schemes. The optimized Runge-Kutta 6-stage stencil (Bogey and Bailly 2004) presents a large stability bandwidth.

Table 3 Comparison of the stability bounds of the Bogey's 6-stage Runge Kutta scheme with the classical Taylor 5-stage scheme. The table reports the upper bound values on $\omega\Delta t$ ensuring $|G| < 1$. As expected, the optimized Runge-Kutta scheme (Bogey 6-stage) presents a large stability bandwidth and therefore is well adapted to achieve an optimal accuracy

	Taylor 5-stage	Bogey 6-stage
$\omega\Delta t <$	1.1	4.0

to consider the geological model as a union of homogeneous sub-domains. The approximation of the stability condition is then performed at each grid point.

Assuming constant values of matrices A_ξ and A_η , the spatial Fourier transform of S reads

$$\hat{S}(\hat{U}) = (iA_\xi k_\xi + iA_\eta k_\eta)\hat{U}. \quad (22)$$

The spectrum of $\hat{S}$ is the spectrum of the matrix $(iA_\xi k_\xi + iA_\eta k_\eta)$. On the other hand, the spectrum of $(iA_\xi k_\xi + iA_\eta k_\eta)$, denoted sp , is

$$sp = \left\{ \pm v_p \sqrt{(ak_\eta + k_\xi)^2 + (bk_\eta)^2}, \pm v_s \sqrt{(ak_\eta + k_\xi)^2 + (bk_\eta)^2}, 0 \right\}, \quad (23)$$

with (a, b) the metrics of the coordinate transformation, v_p is the P-wave velocity and v_s accounts for the S-wave velocity.

The spatial differentiation is accurate up to a tolerance threshold on the group velocity reported in Table 1. Assuming that $\Delta\xi = \Delta\eta = \Delta x$, we obtain

$$|\Lambda_{max}| < c \frac{v_p}{\Delta x} \sqrt{(1 + |a|)^2 + b^2}. \quad (24)$$

Table 4 Values of Ξ for different combinations of FD11o/Runge-Kutta schemes. The error on spatial differentiation is allowed up to 5×10^{-3} . The combination of two optimized stencils in both time and space offers the best accuracy

	Taylor 5-stage	Bogey 6-stage
Ξ (FD11o)	0.74	2.70

We deduce then the general form of the stability condition in curvilinear coordinates

$$\frac{v_p \Delta t}{\Delta x} \sqrt{(1 + |a|)^2 + b^2} < \Xi. \quad (25)$$

Here, Ξ is a constant value that depends on the discrete scheme. As expected, the condition depends on the geometry of the topography: the steeper the topography, the more unstable is the scheme. From a numerical viewpoint, a steep topography squeezes the grid cells, which in turn induces instability of the curvilinear implementation. In Table 4 we report the values of Ξ obtained for the FD11o stencil used with the two different Runge-Kutta schemes illustrated in Fig. 10. The discrete schemes (FD11o/Bogey 6-stage) offer the largest bandwidth of stability.

In case of flat topography, the relation (25) reduces to the well-known condition

$$\frac{v_p \Delta t}{\Delta x} \sqrt{2} < \Xi. \quad (26)$$

The analysis of the stability condition given above is somehow rough and approximate. Although, Ganzha and Vorozhtsov (2001) applied the Von Neumann analysis to the fluid dynamics problem in curvilinear coordinates, the methodology is not fully appropriated; the matrices A_ξ and A_η not only include the elastic parameters ρ , λ and μ but also the metrics of the coordinate transformation. Assuming a constant Jacobian matrix is physically incoherent.

An appropriate method to tackle the stability issues would probably be the energy method. Its advantage lies in its ability to handle matrices with variable coefficients. For instance, Levy and Tadmor (1998) analysed the stability of multi-stage Runge-Kutta schemes (from one to four stages) for the convection diffusion problem, where the matrices are symmetric. However, in our case the matrices A_ξ and A_η do not have a nice structure nor nice properties. Thus, this analytical analysis seems to be untractable even for the Cartesian case.

For these reasons and assuming that only P-waves have to be taken into account, since they have the highest velocities, we studied in the Appendix B the stability of the acoustic wave

equation in a more rigorous way. We established a similar condition as in equation (25).

NUMERICAL VALIDATION

This section is devoted to numerical validation of the curvilinear scheme on 2D academic test problems that exhibit difficult features encountered in real life applications. We first illustrate the numerical behaviour of the approach on a flat topography where the curvilinear scheme reduces to a classical Cartesian solver. Then, we investigate a case of strong contrast on Poisson's ratio. Finally, we study Gaussian hill topography. In order to assess the numerical validity of the simulations, we display both snapshots and seismograms; we also make comparisons with analytical solution when available or with solutions computed with a different approach suited for these simulations namely the Cartesian staggered grid formulation and the spectral finite element method implemented within SpecFEM code (Komatitsch and Vilotte 1998).

For all the numerical experiments, the seismic source is a moment source approximation and the source time function $f(t)$ is a Ricker wavelet defined by:

$$f(t) = (2\pi^2 f_c^2 (t - t_0)^2 - 1) \exp(-\pi^2 f_c^2 (t - t_0)^2), \quad (27)$$

where f_c denotes the central frequency. For all the following tests a central frequency of 20 Hz is considered, which corresponds approximately to a maximum frequency of 50 Hz.

Case of flat topography

In order to assess the accuracy of the non-centred stencils at the vicinity of the free surface, we performed a numerical experiment on a homogeneous model. The total size of the model is 2 km per 1.9 km. A grid cell size of 5 m is selected, which results in 400×380 grid cells. The P-wave velocity is 4000 m/s, the S-wave velocity is 2800 m/s and the density is equal to 1500 kg/m³. The source and the receiver are located

at depth $z = 10$ m, whereas the offset source-receiver is equal to 700 m.

The time step of the simulation is equal to 0.5×10^{-3} s, which results in a stability condition equal to 0.81. The Rayleigh wave velocity is equal to 2651 m/s, which corresponds to 10.6 points per minimal Rayleigh wavelength.

In Fig. 11 we illustrate the propagation of the v_z component, whereas Figs 12 and 13 depict a comparison between the simulated waves and those obtained by the well known Cagniard-de Hoop method (Garvin 1956). The wave modes are the Rayleigh wave, the direct P-wave with ghost, the reflected PS wave and the S^* (Gutowski *et al.* 1984; Mittet 2002). It can be seen that computed and analytical seismograms show a very good agreement.

Case of strong heterogeneity

In this section we investigate the numerical behaviour of the conventional grid in presence of strong heterogeneities. The

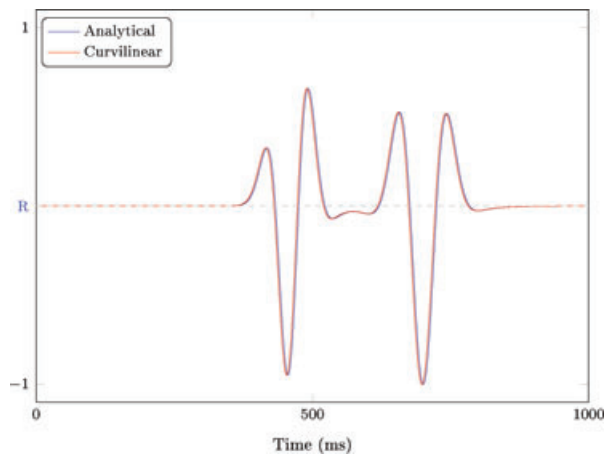


Figure 12 Recorded v_x component at receiver R (red colour) the flat topography model, compared to the analytical result (blue colour). The first arrival are the P-wave and the second arrival is the Rayleigh wave. The overlap between the two seismograms is fairly good.

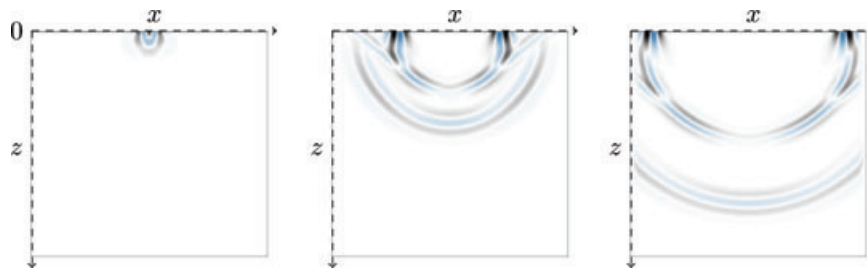


Figure 11 Snapshots of v_z propagation on a flat topography model. The wave modes are the Rayleigh wave, the direct P-wave with ghost, the reflected PS wave and the S^* .

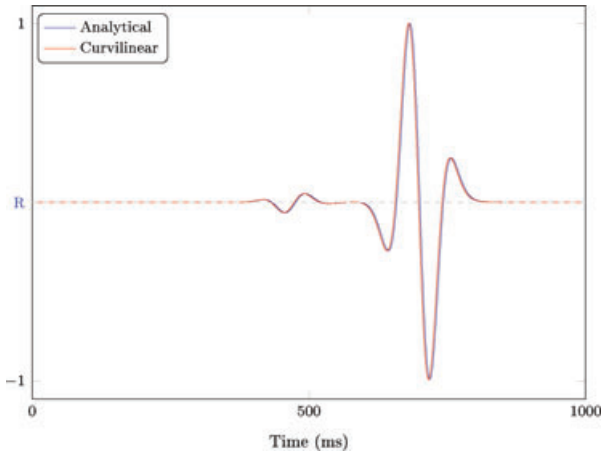


Figure 13 Recorded v_z component at receiver R (red colour) for the flat topography model, compared to the analytical result (blue colour). The first arrivals are the P-wave and the second arrival is the Rayleigh wave. The two seismograms show a very good agreement.

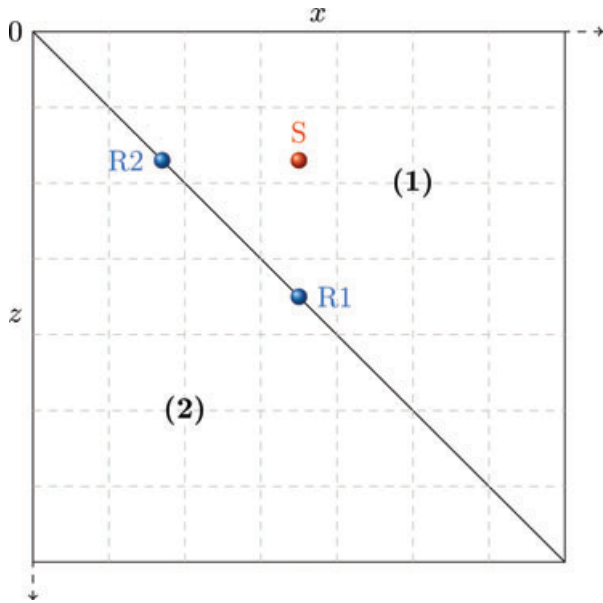


Figure 14 The input model includes two layers. The source S is located at (1000 m, 500 m) while the receivers (R1, R2) are set at (1000 m, 1000 m) and (500 m, 500 m).

model consists of two adjacent layers separated by a 45 degree dipping interface so that the discontinuity is well described by the mesh. The total size of the model is 2 km by 2 km. A grid cell size of 5 m is considered, which results in 400×400 grid cells. In Fig. 14 we depict a sketch of the model, while in Table 5 we report the elastic properties of its different layers.

The receivers are located on the dipping interface at the (2) medium, in order to study the effect of the strong discontinu-

Table 5 Elastic properties of the two layers (1) and (2). v_p is the P-wave velocity, v_s is the S-wave velocity, ρ is the density and ν is the Poisson's ratio

	v_p (m/s)	v_s (m/s)	ρ (kg/m ³)	ν
(1)	3000	1875	1500	0.179
(2)	4500	2250	2000	0.333

ity. To be complete, we compare our results against a classical high-order finite difference scheme on a staggered grid (8th order in space and 4th in time).

Figure 15 reports the snapshots on the v_x component while Figs 16–19 study the recorded seismograms. As shown, the agreement between the two simulations is clearly good.

Case of Gaussian hill topography

In this case, the topography is defined by

$$\begin{cases} z_0(x) = -500 \left(\exp\left(\frac{(x-x_0)^2}{q}\right) - \exp\left(\frac{-x_0^2}{q}\right) \right), \\ q = 10^5, \\ x_0 = 1000\text{m}. \end{cases} \quad (28)$$

The total size of the model is 2 km per 1.9 km. The grid step is equal to 5 m, yielding 400×380 grid cells. The P-wave velocity is equal to 3500 m/s, the S-wave velocity is 2000 m/s and the density is 1500 kg/m³. Two receivers are considered for this simulation, the first one is located at ($x = 500$ m, $z = 214.392$ m), while the second one is at ($x = 750$ m, $z = 214.392$ m). The position of the source within the model is ($x = 1000$ m, $z = 131.943$ m) (see Fig. 20).

We consider a time step of 0.5×10^{-3} s. As in the previous case, the Cartesian stability condition is 0.63 and the number of points per minimal S-wavelength is equal to 8.

In Fig. 21 we illustrate the propagation of the velocity components. We further compare the recorded seismograms to those computed by the spectral element approach (see Fig. 22). The spectral element method and the curvilinear approach do not describe the topography geometry in exactly the same manner. Furthermore, the source and the receiver positions are difficult to locate at the same position for the two codes as the meshes differ. These differences may explain the observed slight mismatches between the computed results that might be observed on this example.

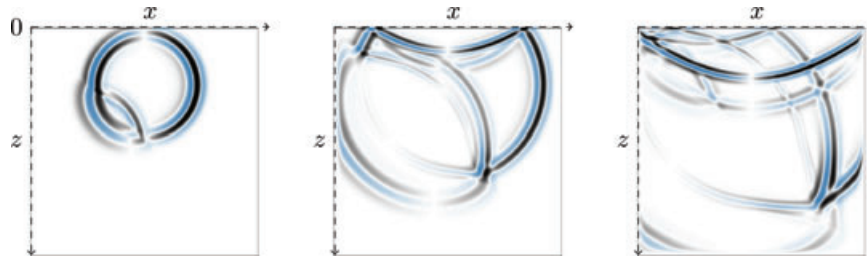


Figure 15 Snapshots on the horizontal velocity component v_x using our curvilinear scheme on conventional grid.

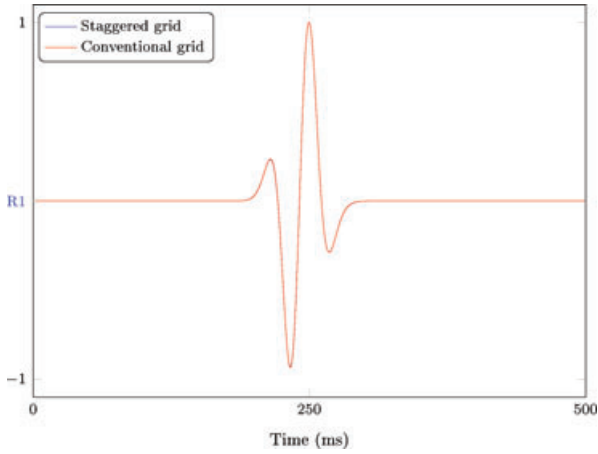


Figure 16 Comparison between the recorded v_x component at the R1 position using our curvilinear scheme (red), against the seismogram obtained with a high-order finite difference method on a staggered grid (blue).

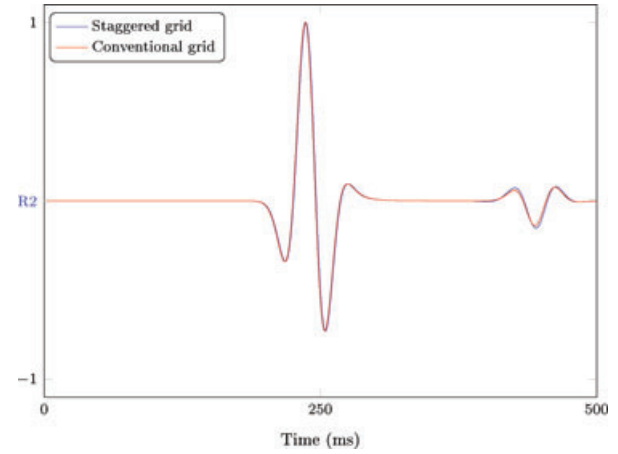


Figure 18 Comparison between the recorded v_x component at the R2 position using our curvilinear scheme (red), against the seismogram obtained with a high-order finite difference method on a staggered grid (blue).

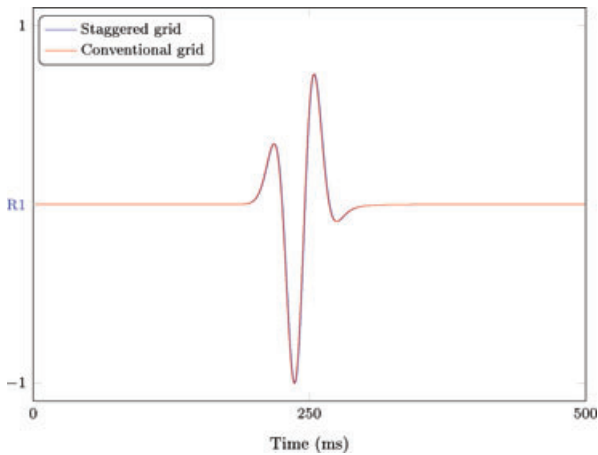


Figure 17 Comparison between the recorded v_z component at the R2 position using our curvilinear scheme (red), against the seismogram obtained with a high-order finite difference method on a staggered grid (blue).

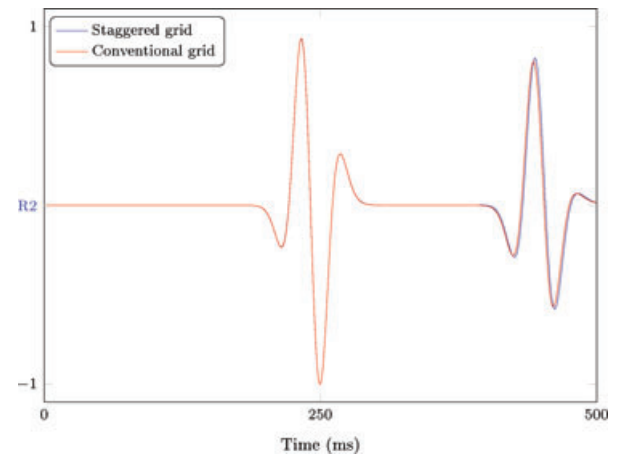


Figure 19 Comparison between the recorded v_z component at the R1 position using our curvilinear scheme (red), against the seismogram obtained with a high-order finite difference method on a staggered grid (blue).

CONCLUSIONS

In this work we studied and developed an efficient scheme to handle elastic wave simulation in presence of topography. Foremost, we recalled the modified equations and the boundary conditions on the continuous problem including the free surface condition. Then, we used optimized stencils and optimized selective filters adapted from aeroacoustics. The use of a conventional grid allowed us to directly extend the non-centred stencils at the boundaries to our problem in curvilinear coordinates. The stability and dispersion theoretical analysis of the scheme was also performed. We further exhibited a

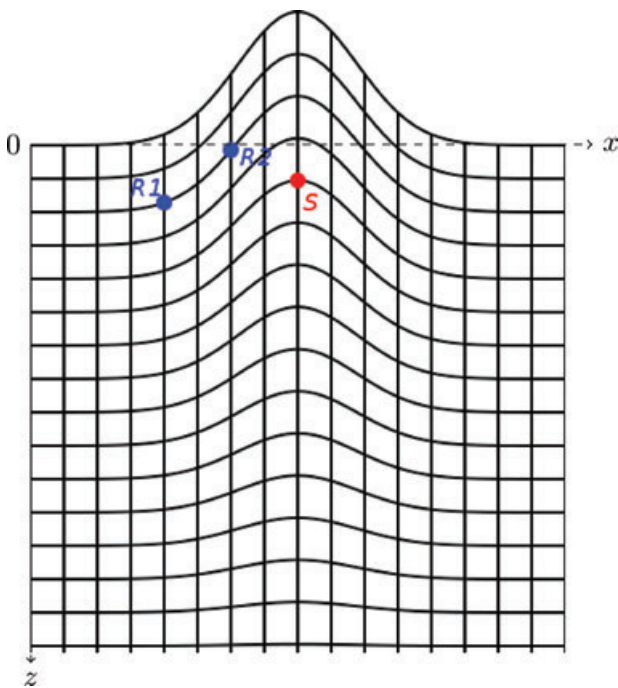


Figure 20 Gaussian hill model. The red color indicates the source position whereas the blue color shows the two receivers locations.

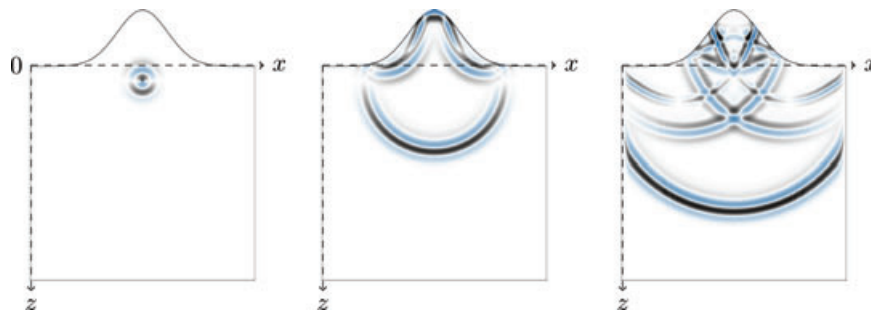


Figure 21 Snapshots of v_z resulting on the propagation of elastic waves on a Gaussian hill model. The snapshots are recorded at time 100×10^{-3} s, 250×10^{-3} s and 400×10^{-3} s.

new stability condition involving the metrics of the coordinate transformation.

We carried out numerical experiments in order to investigate and assess the accuracy of the free surface implementation. For that, we tested our curvilinear scheme on 2D academic problems and compared the results against analytical solutions when available or against numerical results using the SpecFEM package in case of curved topography. We also tested the ability of our grid to handle strong heterogeneities on Poisson's ratio by comparing the results with a classical finite difference scheme on the staggered grids.

By construction, the curvilinear finite difference solver cannot handle a discontinuity on the free surface. Nevertheless, it constitutes an alternative to more powerful methods like spectral element and discontinuous Galerkin methods in most of the cases encountered in industrial applications, thanks to its effective parallel performance and scalability as well as its easy model construction.

The method can be straightforwardly extended to elastic wave propagation in 3D isotropic media. Moreover, the method can be applied to viscoelastic and anisotropic equations. The modified equations are readily deduced according to the chain rule presented previously. Additionally, we believe that curvilinear coordinates could be adapted to implement a fine-grained mesh near the boundaries and more generally to fit complex shapes where staircase approximation of Cartesian finite differences introduces spurious diffractions (e.g., salt diapirs).

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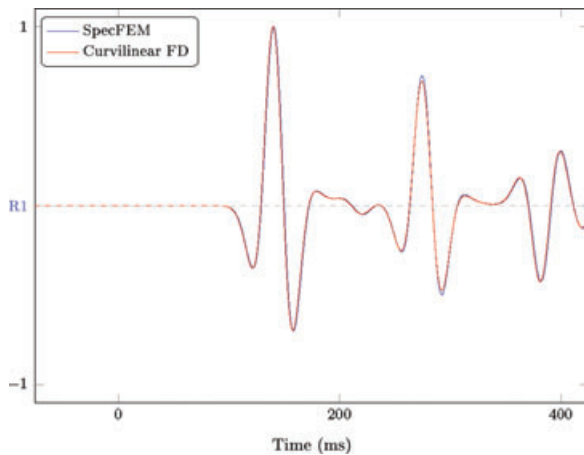


Figure 22 Comparison between the seismogram obtained with the curvilinear code (red) and that simulated with the spectral element approach (blue) for the v_x component on the Gaussian hill model. The considered receiver is R1.

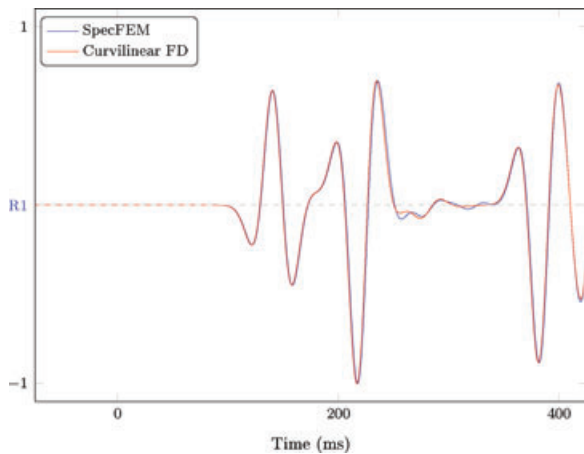


Figure 23 Comparison between the seismogram obtained with the curvilinear code (red) and that simulated with the spectral element approach (blue) for the v_z component on the Gaussian hill model. The considered receiver is R1.

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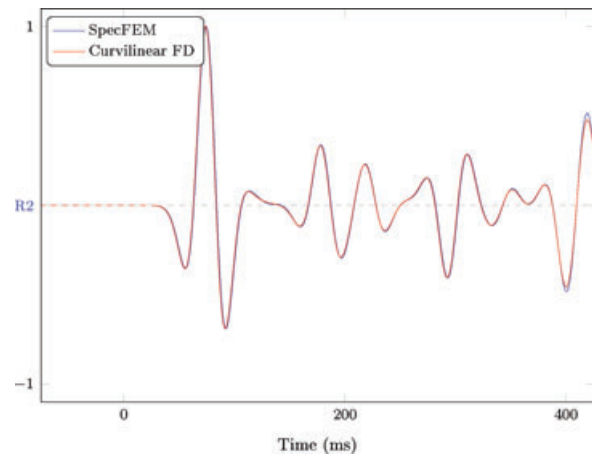


Figure 24 Comparison between the seismogram obtained with the curvilinear code (red) and that simulated with the spectral element approach (blue) for the v_x component on the Gaussian hill model. The considered receiver is R2.

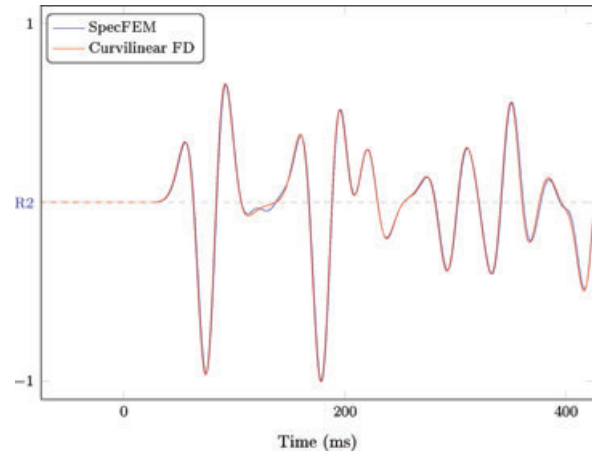


Figure 25 Comparison between the seismogram obtained with the curvilinear code (red) and that simulated with the spectral element approach (blue) for the v_z component on the Gaussian hill model. The considered receiver is R2.

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APPENDIX A: FREE SURFACE CONDITION

In the (ξ, η) basis, the normal vector to horizontal grid lines is defined by

$$\vec{n}(\xi, \eta) = \frac{1}{\sqrt{a^2(\xi, \eta) + b^2(\xi)}} \begin{pmatrix} a(\xi, \eta) \\ b(\xi) \end{pmatrix}. \quad (\text{A1})$$

The continuity of the stress tensor on the top of the geological model implies the free surface condition. Thus, we have

$$\sigma(\xi, 0, t) \cdot \vec{n}(\xi, 0) = 0. \quad (\text{A2})$$

Following Hestholm and Ruud (1994) and applying Hooke's law in isotropic media, we get the free surface condition in curvilinear coordinates:

$$\begin{cases} \frac{\partial v_x}{\partial \eta}(\xi, 0, t) = \Gamma_x^x(\xi) \frac{\partial v_x}{\partial \xi}(\xi, 0, t) + \Gamma_x^z(\xi) \frac{\partial v_z}{\partial \xi}(\xi, 0, t), \\ \frac{\partial v_z}{\partial \eta}(\xi, 0, t) = \Gamma_z^x(\xi) \frac{\partial v_x}{\partial \xi}(\xi, 0, t) + \Gamma_z^z(\xi) \frac{\partial v_z}{\partial \eta}(\xi, 0, t), \end{cases} \quad (\text{A3})$$

where Γ_x^x , Γ_x^z , Γ_z^x and Γ_z^z are defined by

$$\begin{cases} K(\xi) = b(\xi)(1 + z_0'^2(\xi))^2(\lambda + 2\mu)(\xi, 0), \\ \Gamma_x^x(\xi) = K^{-1}(\xi)z_0'(\xi)((3\lambda + 4\mu)(\xi, 0) + z_0'^2(\xi)(\lambda + 2\mu)(\xi, 0)), \\ \Gamma_x^z(\xi) = K^{-1}(\xi)(-\lambda + 2\mu)(\xi, 0) + z_0'^2(\xi)\lambda(\xi, 0), \\ \Gamma_z^x(\xi) = K^{-1}(\xi)(-\lambda(\xi, 0) + z_0'^2(\xi)(\lambda + 2\mu)(\xi, 0)), \\ \Gamma_z^z(\xi) = K^{-1}(\xi)z_0'(\xi)(-\lambda(\xi, 0) + z_0'^2(\xi)(\lambda + 2\mu)(\xi, 0)). \end{cases} \quad (\text{A4})$$

A formulation of the free surface condition was first published by Hestholm and Ruud (1994), their formulation seems to present two singularities in $|z_0'(\xi)| = (\lambda + 2\mu/\lambda)^{1/2}(\xi, 0)$ and may be unstable or inaccurate (Robertsson 1996). Later on Hestholm and Ruud (2000) (equations (30) and (31)) in

their paper) published a correct expression of the free surface condition.

We notice that in the case of flat topography $z_0(\xi) = 0$ equation (A3) reduces to equation (A5), which is the classical form of the free surface conditions in Cartesian coordinates:

$$\begin{cases} \frac{\partial v_x}{\partial v_z}(x, 0) = -\frac{\partial v_z}{\partial x}(x, 0), \\ \frac{\partial v_z}{\partial z}(x, 0) = -\frac{\lambda(x, 0)}{\lambda(x, 0) + 2\mu(x, 0)} \frac{\partial v_x}{\partial x}(x, 0). \end{cases} \quad (\text{A5})$$

When $(z_0' \rightarrow \pm\infty)$, the normal vector $\vec{n}(\xi, 0)$ on the topography tends to

$$\vec{n}(\xi, 0) \rightarrow \begin{pmatrix} \pm 1 \\ 0 \end{pmatrix}. \quad (\text{A6})$$

The free surface condition is then written

$$\begin{cases} \sigma_{xx} = 0, \\ \sigma_{xz} = 0. \end{cases} \quad (\text{A7})$$

The expression of the free surface developed in equation (A3) is no longer adapted for our modelling. Therefore, we cannot handle a very steep geometry where the dip angle of the topography increases too much.

APPENDIX B: STABILITY CONDITION FOR THE SCALAR WAVE EQUATION

We derive hereunder the stability condition for the 2D acoustic wave equation in curvilinear coordinates:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(\xi, \eta, t) = c^2(\xi, \eta) \Delta_c u(\xi, \eta, t) = c^2(\xi, \eta) (D_\xi^2 + D_\eta^2) u(\xi, \eta, t), \\ D_\xi = \frac{\partial}{\partial \xi} + a \frac{\partial}{\partial \eta}, \\ D_\eta = b \frac{\partial}{\partial \eta}, \end{cases} \quad (\text{B1})$$

with c the velocity of the medium.

Because of the non trivial expression of Δ_c , we cannot use a plane wave decomposition $(\omega - k)$ to study the stability of this equation. Thus we study the spectrum $sp(\tilde{\Delta}_c)$ of the discretized curvilinear Laplacian $\tilde{\Delta}_c$, which is a negative definite operator. The largest eigenvalue $-\Lambda^2$ in absolute value of the Laplacian corresponds to the highest frequency mode, which is the most unstable. Λ can be interpreted physically as a characteristic wave-number (inverse wavelength) of its associated vibration mode. In terms of this eigenvalue, the stability condition can be expressed after application of time Fourier transform in the case of a 2nd-order Taylor scheme, we have

Table B1 Stability bounds for the acoustic wave equation in curvilinear coordinates

Time order	2	4	6	8
$\delta <$	1	3.4640	2.7510	4.6340

$$0 \leq \sin^2 \frac{\omega \Delta t}{2} = \Lambda^2 \frac{c^2 \Delta t^2}{4} \leq 1, \quad -\Lambda^2 \in sp(\tilde{\Delta}_c). \quad (B2)$$

For an arbitrary choice of topography, the spectrum of the Laplacian cannot be computed explicitly. Only an upper-bound of the eigenvalues can be derived.

Noting that

$$\rho(\Delta_c) \leq \rho(D_\xi^2) + \rho(D_\eta^2) \leq \rho(D_\xi^2) + \rho(D_\eta)^2, \quad (B3)$$

with $\rho(A) = \max_i |\lambda_i|$, $\lambda_i \in sp(A)$.

We assume a regular grid spacing $\Delta x = \Delta \xi = \Delta \eta$ and a $2N + 1$ -point centred stencil. We apply the Gershgorin circle theorem in order to bound the eigenvalues of the discrete Laplacian. Thus we have

$$\begin{cases} \Lambda^2 \leq \frac{4}{\Delta x^2} \left(\sum_{j=1}^N |a_j| \right)^2 \left((1 + |a|)^2 + b^2 \right), \\ -\Lambda^2 \in sp(\tilde{\Delta}_c). \end{cases} \quad (B4)$$

Here, $\sum_{j=1}^N |a_j|$ is the sum of the coefficients of the spatial stencil.

By combination of equations (B2) and (B4), we obtain the sufficient/simplified stability condition

$$\delta = \frac{c^2 \Delta t^2}{\Delta x^2} \left(\sum_{j=1}^N |a_j| \right)^2 \left((1 + |a|)^2 + b^2 \right) \leq 1. \quad (B5)$$

In the case of a flat topography, this stability condition reduces to the well-known relation

$$\frac{c \Delta t}{\Delta x} \sqrt{2} \left(\sum_{j=1}^N |a_j| \right) \leq 1. \quad (B6)$$

Performing a similar spectral analysis, we can extend this relationship to higher-order schemes in time (N_t) based on the Lax-Wendroff procedure (Dablain 1986). Indeed, the eigenvalues $-\Lambda^2$ of $\tilde{\Delta}_c$ must satisfy the following inequality

$$0 \leq \sin^2 \frac{\omega \Delta t}{2} = -\frac{1}{2} \sum_{k=1}^{N_t/2} (-1)^k \frac{c^{2k} \Delta t^{2k}}{(2k)!} \Lambda^{2k} \leq 1. \quad (B7)$$

Table B1 reports the numerical bounds of the stability condition for higher order in time.

APPENDIX C: FINITE DIFFERENCE COEFFICIENTS

For a centred $(2N+1)$ -point stencil with antisymmetric coefficients a_j , the approximation of the first derivative of a function f reads

$$\frac{\tilde{\partial} f}{\partial x}(x_0) = \frac{1}{\Delta x} \sum_{j=1}^N a_j (f(x_0 + j \Delta x) - f(x_0 - j \Delta x)), \quad (C1)$$

a_j for the different studied stencils are defined in Table C1.

The non-centred derivative of a function f using M points left and N points right writes

$$\frac{\tilde{\partial} f}{\partial x}(x_0) = \frac{1}{\Delta x} \sum_{j=-M}^N a_j f(x_0 + j \Delta x). \quad (C2)$$

The finite difference coefficients of the Bogey 11-point non-centred stencils are reported in Table C2.

The Runge-Kutta algorithm for time differencing writes

$$U^{n+1} = U^n + \sum_{j=1}^p \gamma_j (\Delta t)^j \frac{\partial^j U^n}{\partial t^j}. \quad (C3)$$

The coefficients γ_j for the two studied stencils are reported in Table C3.

Table C1 Finite difference coefficients for the centered stencils discussed in this work

	FD7s	FD7drp	FD9s	FD9o	FD11s	FD11o
a_5	0	0	0	0	0.00079365	0.00248459
a_4	0	0	-0.00357142	-0.00765090	-0.00992063	-0.02077940
a_3	0.01666666	0.02084314	0.03809523	0.05946358	0.05952381	0.09032000
a_2	-0.15000000	-0.16670590	-0.20000000	-0.24467863	-0.23809523	-0.28651117
a_1	0.75000000	0.02084314	0.80000000	0.84157012	0.83333333	0.87275699

Table C2 Non-centered finite difference coefficients used for free surface implementation

FD46o	FD37o	FD28o	FD19o	FD010o
0.016756572	−0.013277273	0.057982271	−0.180022054	−2.391602219
−0.117478455	0.115976072	−0.536135360	−1.237550583	5.832490322
0.411034935	−0.617479187	−0.264089549	2.484731692	−7.650218001
−1.130286765	−0.274113948	0.917445878	−1.810320814	7.907810563
0.341435872	1.086208765	−0.169688364	1.112990048	−5.922599052
0.556396830	−0.402951626	−0.029716326	−0.481086916	3.071037015
−0.082525734	0.131066986	0.029681617	0.126598690	−1.014956769
0.003565834	−0.028154858	−0.005222483	−0.015510730	0.170022256
0.001173034	0.002596328	−0.000118806	0.000018609	0.002819958
−0.000071773	0.000128743	−0.000118806	0.000156448	−0.004791009
−0.000000352	0.000000000	−0.000020069	−0.000007390	−0.000013063

Table C3 Runge-Kutta coefficients used within this work

	Taylor 5-stage	Bogey 6-stage
γ_1	1	1
γ_2	1/2	1/2
γ_3	1/3	0.1659197713678104
γ_4	1/4	$4.091973204095216 \times 10^{-2}$
γ_5	1/5	$7.555704391128985 \times 10^{-3}$
γ_6		$8.914212608018190 \times 10^{-4}$